

Incident Report

Executive Summary

A simulated Security Operations Center (SOC) environment was implemented using Wazuh SIEM, Kali Linux, Metasploitable2, and TheHive. The objective of the project was to simulate attack activity, monitor security events, perform threat hunting, and demonstrate incident response procedures. Security monitoring operations generated alerts related to CIS benchmark compliance failures and network monitoring activities. Threat hunting and event analysis were conducted using Wazuh dashboards and endpoint telemetry. MITRE ATT&CK techniques including T1046 (Network Service Scanning), T1078 (Valid Accounts), and T1210 (Exploitation of Remote Services) were mapped to observed activities.

Timeline

Time	Activity
20:00	Wazuh SIEM environment initialized
20:10	Wazuh agent connected successfully
20:15	Threat hunting activities initiated
20:18	Security alerts generated
20:25	Metasploit attack simulation executed
20:35	Event analysis and triage completed
20:45	Incident response workflow documented

Root Cause Analysis

The primary cause of generated alerts was weak system configuration and CIS benchmark non-compliance. Security monitoring rules identified abnormal system activity and endpoint configuration issues. Limited endpoint telemetry reduced visibility into advanced command execution monitoring, highlighting the need for enhanced logging and automated response capabilities.

Recommendations

- Improve CIS benchmark compliance
- Reduce exposed services and insecure configurations
- Implement advanced endpoint logging
- Enhance automated incident response workflows
- Improve threat intelligence correlation
- Conduct regular vulnerability assessments and monitoring